



Technical Note

Note on the sequence dependent setup scheduling problem (SDSSP) with earliness and tardiness penalties



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ARTICLE INFO

Article history:

Received 3 July 2014

Received in revised form 25 August 2014

Accepted 23 October 2014

Available online 29 October 2014

Keywords:

Sequence dependent

Setup scheduling problem

Earliness and tardiness

ABSTRACT

In the paper: "Minimizing the Sum of Earliness/Tardiness in Multi-machine Scheduling: A Mixed Integer Programming Approach" by Zhiwei Zhu and Ronald B. Heady, published in Computers & Industrial Engineering 38 (2000) 297–305, they present two formulations. Both their triple index and double index mathematical formulations address the sequence dependent setup scheduling problem (SDSSP) with earliness and tardiness penalties. These formulations were compared to the network flow based formulation of Anderson et al. in the paper: "An Efficient Network-Based Formulation for Sequence Dependent Setup Scheduling on Parallel Identical Machines", published in Mathematical and Computer Modeling, Volume 57, Issues 3–4, February 2013, p. 483–493. Through this comparative analysis, an omission in the double index Zhu and Heady formulation was discovered.

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In their paper, Zhu and Heady (2000) present two formulations. Both their triple index and double index mathematical formulations address the sequence dependent setup scheduling problem (SDSSP) with earliness and tardiness penalties. These two formulations were compared to the network flow based formulation of Anderson, Blocher, Bretthauer, and Venkataramanan (2013) and the algorithm of Balakrishnan, Kanet, and Sridharan (1999) in the paper by Anderson et al. (2013). Through this comparative analysis, an omission in the double index Zhu and Heady formulation was discovered.

The Zhu and Heady nomenclature and double index formulation is defined as follows.

B	a large number
d_i	due date for job i
e_i	earliness cost per period for job i
E_i	length of time job i is early
p_{im}	processing time for job i using machine m
M	number of machines available
N	number of jobs to be scheduled
s_{ji}	setup time for job i when it immediately follows job j
s_{0i}	setup time for job i when it is the first in the queue
t_i	tardiness cost per period for job i

T_i	length of time job i is tardy
X_i	completion time of job i
Y_{ij}	1 if job i precedes job j , 0 otherwise
Z_{im}	1 if job i is done on machine m , 0 otherwise

Objective Function:

$$\min \sum_{i=1}^N (e_i E_i + t_i T_i)$$

Subject to:

$$X_i - T_i + E_i = d_i \quad i = 1 \dots N \quad (1)$$

$$\sum_{m=1}^M Z_{im} = 1 \quad i = 1 \dots N \quad (2)$$

$$X_j - X_i - BY_{ij} - BZ_{im} - BZ_{jm} \geq p_{jm} + s_{ji} - 3B$$

$$j \neq i, i = 1 \dots N, j = i + 1 \dots N,$$

$$m = 1 \dots M \quad (5a)$$

$$X_i - X_j + BY_{ij} - BZ_{im} - BZ_{jm} \geq p_{im} + s_{ji} - 2B$$

$$j \neq i, i = 0 \dots N, j = i + 1 \dots N,$$

$$m = 1 \dots M \quad (5b)$$

The concepts behind these constraints is sound, however there are two issues in the implementation of the formulation, as it is

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presented. First, the index range for i in constraints (5a) and (5b) is incorrect and should be switched so that i begins with 0 in constraint (5a) and i begins with 1 in constraint (5b), as follows. The addition of the 'all' symbol is simply this author's preference for clarity.

$$\begin{aligned} X_j - X_i - BY_{ij} - BZ_{im} - BZ_{jm} &\geq p_{jm} + s_{ij} - 3B \\ j &\neq i, \forall i = 0 \dots N, \forall j = i + 1 \dots N, \\ \forall m &= 1 \dots M \end{aligned} \quad (5a)$$

$$\begin{aligned} X_i - X_j + BY_{ij} - BZ_{im} - BZ_{jm} &\geq p_{im} + s_{ji} - 2B \\ j &\neq i, \forall i = 1 \dots N, \forall j = i + 1 \dots N, \\ \forall m &= 1 \dots M \end{aligned} \quad (5b)$$

In addition to the modified ranges for index i , it was found that two new constraints were needed, as shown below.

$$\sum_{m=1}^M Z_{0m} = M \quad (\text{new1})$$

$$\sum_{j=1}^N Y_{0j} = N \quad (\text{new2})$$

Since Z_{im} is a multiplier of B (a large number) in constraints 5a and 5b, constraint 'new 1' ensures the large number is 'established' once for each machine, so the constraints function properly. Constraint 'new 2' ensures the available jobs are considered for the initial position in each sequence. These changes are necessary for successful implementation of the model.

References

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